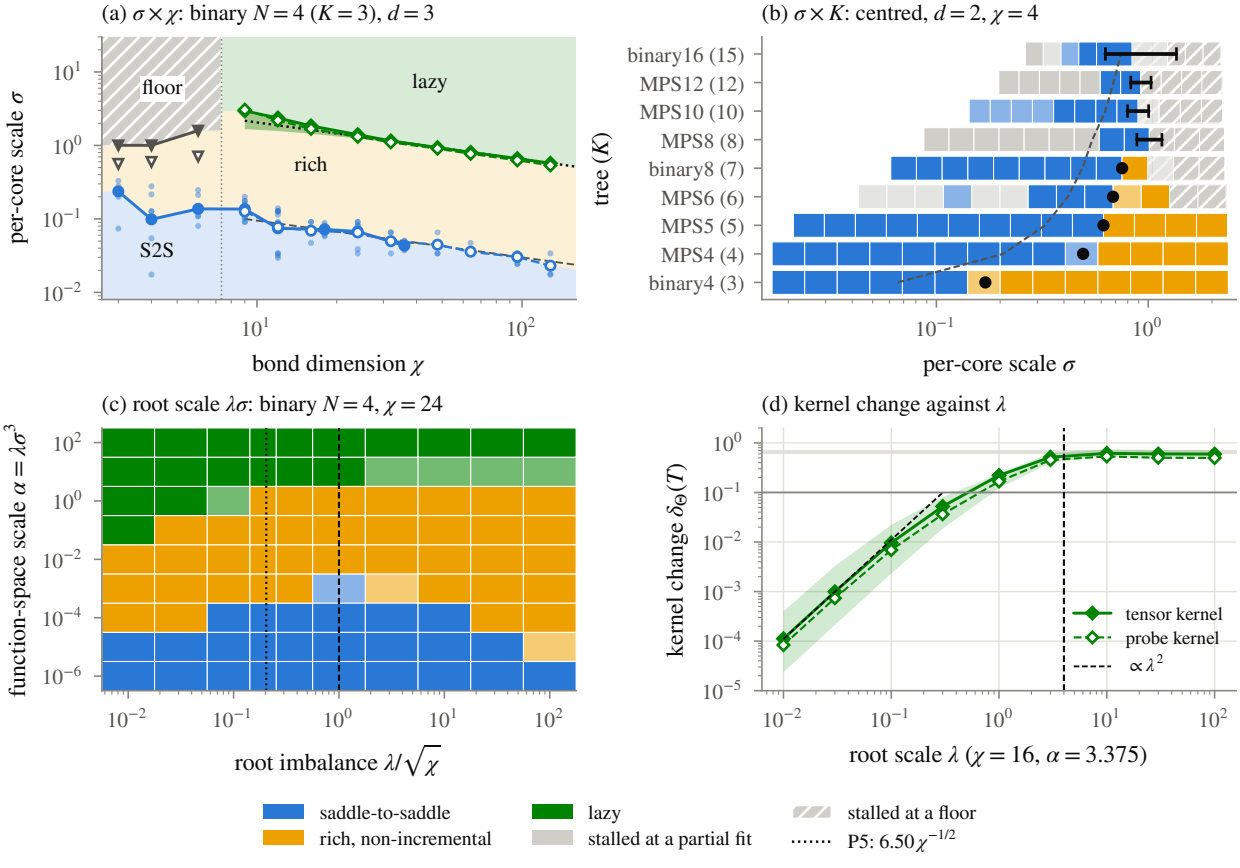
(a) $\sigma \times \chi$: binary $N = 4$ ($K = 3$), $d = 3$
floor
lazy
rich
S2S
per-core scale σ
bond dimension χ
(b) $\sigma \times K$: centred, $d = 2$, $\chi = 4$
binary16 (15)
MPS12 (12)
MPS10 (10)
MPS8 (8)
binary8 (7)
MPS6 (6)
MPS5 (5)
MPS4 (4)
binary4 (3)
tree (K)
per-core scale σ
(c) root scale $\lambda\sigma$: binary $N = 4$, $\chi = 24$
function-space scale $\alpha = \lambda\sigma^3$
root imbalance $\lambda/\sqrt{\chi}$
(d) kernel change against λ
kernel change $\delta_\Theta(T)$
tensor kernel
probe kernel
$\propto \lambda^2$
root scale λ ($\chi = 16$, $\alpha = 3.375$)
saddle-to-saddle
rich, non-incremental
lazy
stalled at a partial fit
stalled at a floor
P5: $6.50\,\chi^{-1/2}$